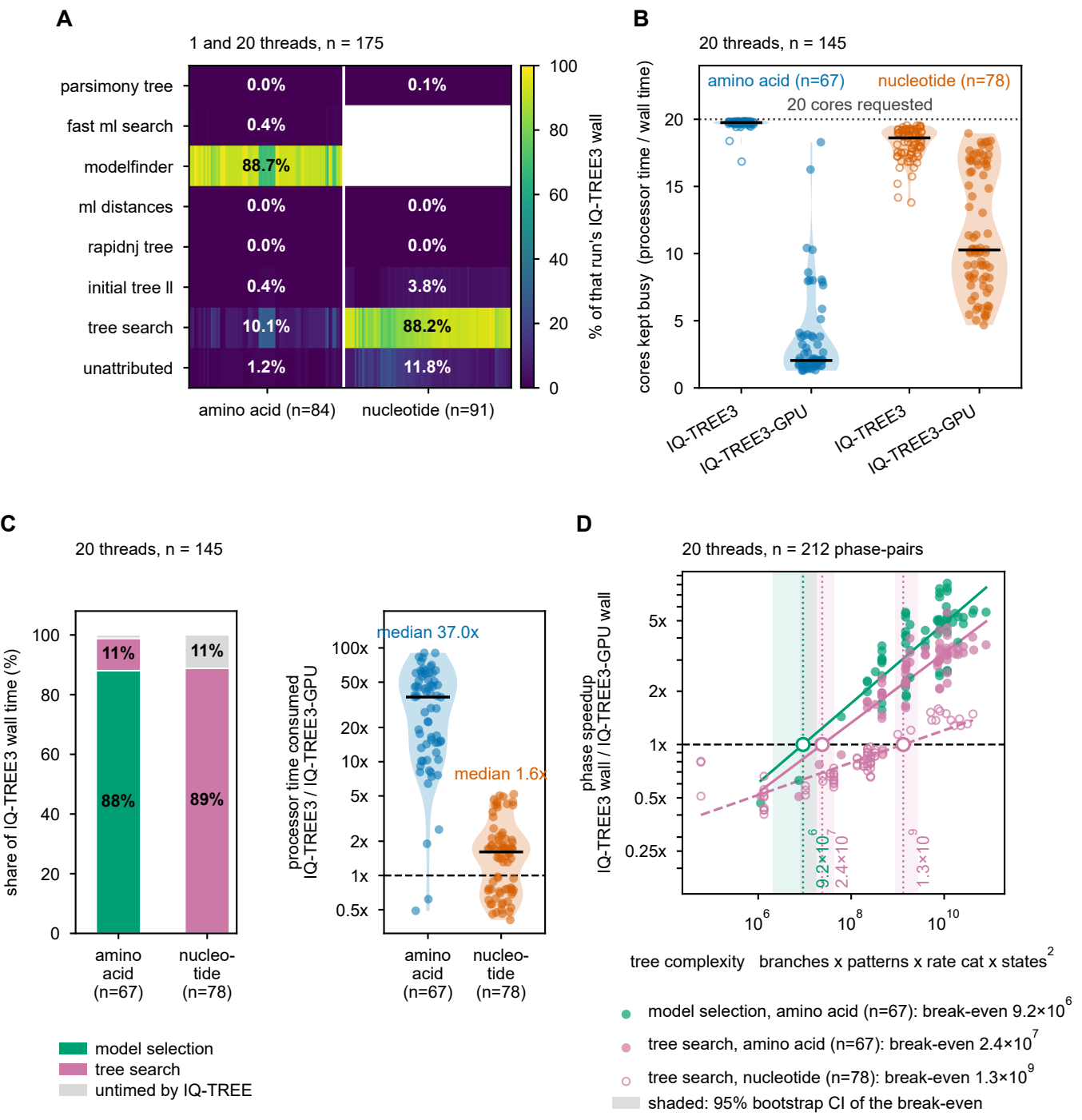

A
1 and 20 threads, n = 175
parsimony tree
fast ml search
modelfinder
ml distances
rapidnj tree
initial tree ll
tree search
unattributed
0.0%
0.1%
0.4%
88.7%
0.0%
0.0%
0.0%
0.0%
0.4%
3.8%
10.1%
88.2%
1.2%
11.8%
amino acid (n=84)
nucleotide (n=91)
100
80
60
40
20
0
% of that run's IQ-TREE3 wall
B
20 threads, n = 145
amino acid (n=67)
nucleotide (n=78)
20 cores requested
cores kept busy (processor time / wall time)
20
15
10
5
0
IQ-TREE3
IQ-TREE3-GPU
IQ-TREE3
IQ-TREE3-GPU
C
20 threads, n = 145
share of IQ-TREE3 wall time (%)
100
80
60
40
20
0
11%
11%
88%
89%
amino acid (n=67)
nucleo-tide (n=78)
model selection
tree search
untimed by IQ-TREE
median 37.0x
median 1.6x
processor time consumed IQ-TREE3 / IQ-TREE3-GPU
100x
50x
20x
10x
5x
2x
1x
0.5x
amino acid (n=67)
nucleo-tide (n=78)
D
20 threads, n = 212 phase-pairs
phase speedup IQ-TREE3 wall / IQ-TREE3-GPU wall
5x
2x
1x
0.5x
0.25x
9.2×10^6
2.4×10^7
1.3×10^9
10^6
10^8
10^10
tree complexity branches x patterns x rate cat x states^2
model selection, amino acid (n=67): break-even 9.2×10^6
tree search, amino acid (n=67): break-even 2.4×10^7
tree search, nucleotide (n=78): break-even 1.3×10^9
shaded: 95% bootstrap CI of the break-even